Journal, Ouro Preto, Brazil, November 2007, VII Meeting of the Southern Hemisphere on Mineral Technology

Saturday 17 November 2007/Sunday 18 November 2007

Another trip to Brazil, to Ouro Preto for a Plenary Lecture at the VII Meeting of the Southern Hemisphere on Mineral Technology. Previous trips to Brazil included a trip to Belo Horizonte (UFMG) in 1991, and a trip with Patricia to Rio de Janeiro (CETEM) in 2000.

Departure from SLC at ~7 am on AA via Chicago, Miami, arriving Sao Paulo Sunday morning. Departure from Sao Paulo to Belo Horizonte, arriving 2 pm Sunday afternoon. Total travel time ~24 hours.

Roberto Galery, Head of Mining Engineering UFMG (Federal University of Minas Gerais) collected us at the airport and arranged for accommodations at the San Diego Convention Hotel. (Papa was last in Belo Horizonte in August/September 1991 to give a short course at UFMG. It was at this time that Maria Azevedo was met. Maria eventually came to the U of U as well as Roberto for graduate study in Papa's research group.). After a swim in the hotel pool and a bit of rest, we toured Belo (Liberty Square and the Serra da Moeda ,Coin Mountain, where the Governor's Estate/Park is located on top of mountain) having pizza Sunday night at an Italian restaurant before returning to the hotel.

Monday 19 November 2007

Had an early morning swim before checking out and going with Roberto to visit UFMG. We had a tour of the facilities and met old friends on the faculty, Antonio Peres, George Valadao (Past Visiting Researcher U of U), Armando Araujo, Afonso Martins (Past Visiting Researcher U of U),
Paulo Brandao, and others. Lunch was taken at a buffet close to the campus. After lunch we reviewed our research on molecular dynamics simulation and let Armando copy the PP presentation for his class. About 4 pm we left Belo arriving in Ouro Preto at ~6 pm. After checking in at the hotel, Pousada Minas Gerais, we eventually went for an authenic Brazilian dinners featuring beans, pork and rice. These were typical Minas Gerais dinners featuring beans. Both dinners had roasted pork loin and rice. In one case the beans were Tutu-mashed beans-and in the other case the dinner had a special name, Feijao de Troupeiro, Trooper beans (mixed) with pork meat,scrambled eggs,and onions/tomatoes I think. The dinners were great.

Tuesday 20 November 2007

Spent all day touring the colonial city Ouro Preto and the Mining and Metallurgy schools at UFOP (Federal University Ouro Preto). Ouro Preto was the old capital of Minas Gerais during the gold rush when the Portugese were colonizing Brazil (18th century). The small city of Ouro Preto is in a mountainous region and consequently there are many hills in the city and everything is paved with cobblestones. Minas Gerais is the mineral rich state with much iron ore and other minerals in addition to gold. I think almost 50% of Brazil's mineral production comes from Minas Gerais.
This region is known for its Imperial Topaz. Such stones are not found in any other geographic location. In fact Roberto and I had a look at some stones in the marketplace this morning.

After the market, we went with Mike Nicol from Murdoch University, formerly at MINTEK in

South Africa, and Guven Onal from Istanbul, ITU, to visit the restored Casa dos Contos, originally Casa dos Contratos (House of Contracts). It seems that this was where the gold was counted and the Portugese Govenor took 1/5 for Portugal. In addition the Casa dos Contratos was used to mint coins and manage the early economy. Eventually the Brazilians got tired of their colonial position. Pedro II and the Portugese soldiers returned to Portugal to contend with Napoleon and Brazil was left to govern itself.

In the afternoon we went to the schools of mining and metallurgy at UFOP and had a tour of facilities. Roberto Galery's brother-in-law runs the SEM and we had a good discussion. The head of the Mining Dept., Carlos gave us topaz samples from the state of Amazonas.

During the Opening Ceremony for the VII Meeting of the Southern Hemisphere on Mineral Technology we had vocal entertainment by the Querubins (Angels), poor children from the streets of Ouro Preto who were organized by a volunteer director for this childrens choir. They were delightful. After the Opening Ceremony, we saw old friends at the reception tonight including Mike Nicol and his wife Hazel (originally from South Africa, they know John and Jean Davidtz) now Australians living in Perth as well as Osseo Asare and his wife Fran from Penn State who have been on leave in Belo Horizonte since August. Also, saw Marisa Monte. She and her husband took us for tennis and lunch (Brazilian Coffee as I recall or maybe the coffee was with Marcello Tavares and his wife) when we were together in Rio de Janeiro some years ago November 2000.

Wednesday 21 November 2007

Most of the day was spent at the conference being held in a restored Parque Metalurgico Centro de Artes e Convencoes, Metallurgical Park for Arfs and Conventions. I seemed to have had some bad food and was miserable most of the day and night. Dinner was at Bene' da Fluta, a wonderful restaurant named after Bene' a self taught street musician who had endeared himself to the city during the last part of the 20th century. Nice paintings of Ouro Preto were presented on the walls of the restaurant including one of Bene'. The food was excellent but I could not eat much. Before retiring I worked on my Plenary Lecture having received words of encouragement from wife, Patricia.

Thursday 22 November 2007

Happy Thanksgiving ! It would be easier to be thankful if I had one of Granny's roasted turkeys with me today. After a rough night, I made it to the convention center. I was introduced by Roberto Galery, former student now professor at UFMG and my Plenary Lecture was presented to a full house. The lecture was well received. After the lecture we went with Roberto Galery and Claudio Scheinder, former student with Professor King at Utah to have lunch at Chafariz (Fountain) Restaurante. For tonight we have a dinner at the Convention Center, the restored Metallurgical Park.

Before dinner, we went to the oldest theater in Ouro Preto, 18th century, to hear a quartet of Brazilian musicians, flute, accordian, guitars, and tambourine, playing some authentic Brazilian music. The theater was typical of such an era with two balconies continuing on all sides with box seats. We had seats on the second level. The evening concluded with a

reception at the conference center and an order of french fries from room service to celebrate Thanksgiving.

Friday 23 November 2007

Today has been quite an interesting day, filled with activities under the direction and organization of Paulo Brandao, UFMG Mining Engineering Faculty. I had been an external member of Paulo's PhD thesis committee at UBC in 1982 I believe. Anyway, Paulo took good care of us today. Unfortunately, my health has taken a turn for the worse and I have had a bit of a relapse. Still having a problem with my digestive system.

We began the day by attending an organ concert in Mariana, a small town ~20 km east of Ouro Preto. As mentioned before, this is the gold region during colonial times and the Portugese prospered from the gold recovered in this mountainous region of Minas Gerais. Apparently there was some competition between Ouro Preto and Mariana. In fact Ouro Preto became the capital for the government and Mariana became the capital for the Catholic church. The unique aspect of this organ concert was that it was being held at the Catedral da Se' de Mariana which has this restored Arp Schnitger organ dating from the 18th century. It was built in North Germany by Arp Schnitger between 1700 and 1710. It was for many years in Portugal at one of the Franciscan churches and subsequently shipped to the first bishop of Mariana in Brazil in 1753 as a gift from the King of Portugal, D.Joao the V. Eventually, after many years of neglect, it was shipped back to Hamburg, Germany for restoration and then returned to Mariana in 1984. As expected the organ bellows were designed to be manually operated and it still functions in that way if desired. The concert was fine with selections from composers, Couperin, Rameau, Bohm, Pachelbel, Corrette, and Balbastre none of whom I have ever heard of.

After lunch at a local restaurant, not sure the food agreed with me, we journeyed to an old gold mine, Mina da Passagem, the oldest gold mine in Minas Gerais from 1817. The opening/entrance is an inclined adit following the gold vein to a depth of 120 meters. We traveled by trolley on rail to examine the mine which closed in 1985 after an estimated 35 tons of gold had been extracted.

Some further comments about the region include the Cantaria, or stone mason trade recently restored by Carlos Pereira who was even invited to Italy to share his techniques with the Italians. I had the opportunity to meet Carlos' son, head of Mining at UFOP, the fellow who gave us some topaz from Amazonas region. The Cantaria was developed for quartzite since there was not suitable marble in Minas Gerais. In addition to the mason work on quartzite, skill in the sculpting of soapstone was developed early on. Soapstone is a metamorphic rock composed chiefly of talc,sericite, I guess. One of the most famous sculptors was Aleijadinho, or “Little Cripple”. He was crippled during the latter years of his life yet was able to do marvelous sculpting of soapstone. They woud tie the tools to his hands since he was not able to grip the tools.

As mentioned Ouro Preto is surrounded by mountains. One peak is very striking for an extruded rock mass with a small satellite towers above the top of the mountain. It is known by its Indian name, the Itacolome peak, which means mother and child. This distinctive feature, Itacolome, is in clear view from the balcony of the hotel room.

Heavy thunderstorms were experienced this afternoon. The steep cobblestone streets of Ouro Preto became so wet that at least one street was closed for fear that cars/vans/trucks would slip on descent and lose control. In fact we had to alter our travel to avoid this street.

Before returning to the hotel we stopped to examine some stones. As you may remember this region around Ouro Preto is the only place where the beautiful Imperial Topaz is found. Stones of great beauty in colors ranging from yellow through sherry to delicate pinks are found. With Paulo's assistance Imperial Topaz stones were purchased for each of the girls; Patricia, Pamela, Jeanette, and Virginia.

Saturday 24 November 2007

Last night we went to have pizza for dinner. Of course I had to have a pizza that was custom designed. Needless to say I had a rough night and thought I would need to go to the doctor. However I was better in the morning and continue to manage with my discomfort. Can't wait to get home for some TLC. There is plenty of Guarana, Goiaba (Guava), and other drinks to wet the palate.

Today was spent mostly on a field trip with Robero Galery to the Samarco iron ore company, a bit further east from Ouro Preto, beyond Mariana. The iron ore is Itabirita and the specular hematite is recovered by flotation of quartz. Most interesting is that the iron concentrate is transported as a thick slurry, 60-70 percent solids, to the coastal port, Ubu, in a 20 inch pipeline, a distance exceeding 100 km. Samarco is one of the important iron ore companies in the iron quadrangle of Minas Gerais. In addition to presentations and tours, Samarco served a late lunch before we returned to Ouro Preto where we spent the remainder of the afternoon enjoying the mineral collection at the Old School of Mines Building, on the city square, which is now a museum.

At the end of the afternoon we met Armando and Mary Arjauo. Armando is on the mining faculty of UFMG. They are quite depressed for the last couple months since they lost their only adopted son 5 years of age to a brain tumor. We had coffee and tried to comfort them. I promised to pray for them. They were really sad and I tried to cheer them. I know Granny would have known what to do. I did the best I could.

Finally some more on the history of Ouro Preto. In the fight for independence the hero/martyr, was Tiradentes who was hung in Ouro Petro by the Portugese as a rebel to demonstrate the power of the Portugese. Ouro Preto was the center of attraction, of course, because of the gold rush. A statue of Tiradentes stands in the city square in Ouro Petro. It seems that Don Joao V or VI returned to Europe to fight Napoleon. His son, Pedro I, took charge in Brazil and eventually declared Brazil to be an independent country. Subsequently his son Pedro II became emperor of this new independent country, Brazil. During these days blacks from Africa were brought as slaves for plantations and for the gold rush. The slaves, when they had a chance, would escape and, because of the treacherous mountain terrain, the Portugese could not find the escapees. In fact the blacks hid and found a small town, Lavras Novas, new mining, which survives to this day, not far from Ouro Preto, with many generations of the black slaves that had been brought from Africa now

enjoying Brazil. Thinking about the slave trade reminds me of
Wilberforce and his life, told in the story, Amazing Grace.

Well, tomorrow we head home with a late night flight out of Sao Paulo.
We will drive to Belo Horizonte after lunch to catch a flight to Sao
Paulo at about 6:30 pm. This visit to Brazil is filled with many memories to add to the memories of visits to Belo in 1991 and to Rio in 2000.

Sunday 25 November 2007

Well, we'll be checking out of Pousada Minas Gerais in about 20 minuites. This morning, after correcting some theses/manuscripts, we went with Roberto to the Ouro Petro Chocolate Shop. Christmas decorations are already in place. We had chocolate drinks prepared with the Ouro Preto chocolate made from cocoa beans grown in the Bahia state. The chocolate is very good.

During conversation with Roberto this morning I learned that most of the Brazilian Indians are on reservations, maintaining their language/culture to a certain extent, no modern conveniences, utilities, etc. The government Indian agency does provide them with help, hospitals, education, etc. In fact the Indians are not considered Brazilian citizens but have their own papers identifying them as Indians. On the other hand, the blacks are quite well integrated, with mixed marriages yielding mulato children, and consequently a complete spectrum of colors exists in Brazilian society with little discrimination regarding color.

On the drive back to Belo Horizonte, we stopped for shopping at a lovely mall and found some Brazilian soccer (futbol) jerseys. After shopping we had lunch at the traditional Churrascu (Barbeque) with sharply dressed waiters passing by the tables with all kinds of meat on skewers ready to slice for you from their skewer at your desire. This was the first time we had gone for food at one of the famous Churrascus, Rodeio grill/restaurant.

Finally, before departing for the airport, we went to Roberto’s home to meet his wife, Fatima and daughter Anna. In addition, Roberto gave us some small industrial grade diamonds for Papa’s diamond collection. These alluvial diamantes (gia mantes) were taken from the Jaquetinhonha River sands near the city of Diamantina, MG, about 400 km northeast of Belo Horizonte.

The GOL flight from Belo left at about 6:30 pm (1:30 pm SLC time), arriving in Sao Paulo at about 7:30 pm. The flight from Sao Paulo on AA departed at midnight with connection through Dallas, Texas, arriving at 11 am Monday morning. A great time was had meeting many Brazilian friends and seeing many sights.